

SOLUTIONS TO SOME OF THE QUESTIONS ON PREVIOUS STATISTICS APPLIED MASTERS COMPREHENSIVE EXAM

These are Solutions that have been found to some of the questions that were asked on the Applied Masters Comprehensive Exams given since 2013 which may be similar to Applied Questions that may be asked on the new Single Masters Comprehensive Exam that is being given from Fall 2019 on. In some cases what is presented here may be the initial questions and solutions that were submitted when the exam was prepared. It is possible that the questions (as well as the solutions that were used for grading) for the exams, which were given, may have changed slightly. Also note that these answers may not be the only way to address the question, in some cases there may be other solutions that would get partial or even full credit.

Also written by each of the questions in Blue are the **COMPETENCIES** that reflect the areas of knowledge that the questions are covering.

SPRING 2019

1. Captain Krik has obtained 5 independent random observations on the magnitude of interstellar blips which have the following values; -7, -4, -1, 4, 8. These are presented as single digits to simplify computation, but assume for your answers these are taken down to two decimal places.

COMPETENCIES – Basic Properties of Distributions and Normal Distribution Methods

- 1 Give the best estimates from this sample for ;the population mean, standard deviation, skewness and kurtosis. Show your formulas and calculations.

$$\text{Mean} = [-7 + -4 + -1 + 4 + 8] / 5 = 0$$

$$\text{Variance} = \{[(-7)^2 + (-4)^2 + (-1)^2 + (4)^2 + (8)^2] - (0)^2\} / (5-1) = 36.5$$

$$\text{Standard Deviation} = \sqrt{36.5} = 6.04$$

$$\begin{aligned}\text{Skewness}^* &= \frac{n\sqrt{n-1}}{n-2} \frac{\sum (X - \bar{x})^3}{\left[\sum (X - \bar{x})^2\right]^{3/2}} = \frac{5\sqrt{5-1}}{5-2} \frac{(-7)^3 + (-4)^3 + (-1)^3 + (4)^3 + (8)^3}{\left[\sum (-7)^2 + (-4)^2 + (-1)^2 + (4)^2 + (8)^2\right]^{3/2}} \\ &= \frac{5\sqrt{4}}{3} \frac{-343 - 64 - 1 + 64 + 512}{\left[\sum 49 + 16 + 1 + 16 + 64\right]^{3/2}} = \frac{10}{3} \frac{168}{1764.12} = 0.317\end{aligned}$$

$$\begin{aligned}\text{Kurtosis}^* &= \frac{n(n+1)(n-1)}{(n-2)(n-3)} \frac{\sum (X - \bar{x})^4}{\left[\sum (X - \bar{x})^2\right]^2} = \frac{5(5+1)(5-1)}{(5-2)(5-3)} \frac{(-7)^4 + (-4)^4 + (-1)^4 + (4)^4 + (8)^4}{\left[\sum (-7)^2 + (-4)^2 + (-1)^2 + (4)^2 + (8)^2\right]^2} \\ &= \hat{b}_R = \frac{120}{6} \frac{2401 + 256 + 1 + 256 + 4096}{\left[49 + 16 + 1 + 16 + 64\right]^2} = \\ &= 20 \frac{7010}{21316} = 6.58 \text{ [Some people will subtract out 3 the mean kurtosis of} \\ &\quad \text{the normal distribution to get 3.58]}\end{aligned}$$

* NOTE – For skewness and kurtosis full credit was given for other small sample adjustments for skewness and kurtosis and substantial credit given for answers that demonstrated understanding of the concept that skewness is estimated by $\frac{\sum [X_i - \bar{x}]^3}{[\text{Variance}]^{3/2}}$ and

that kurtosis is estimated by $\frac{\sum [X_i - \bar{x}]^4}{[\text{Variance}]^2}$

- b, Assume that these five values come from a normal distribution and that the next interstellar blip to occur will be random / independent. Give a 95% confidence interval for the value of this next interstellar blip out to 2 decimal places

General Formula is $\bar{x} \pm t_{n-1, 1-\alpha/2} S \sqrt{(1+1/n)}$ which is $0 \pm t_{4, 1-0.05/2} 6.04 \sqrt{(1+1/5)}$ which is $0 \pm (2.776)(6.04)(1.095)$ or -18.36, 18.36

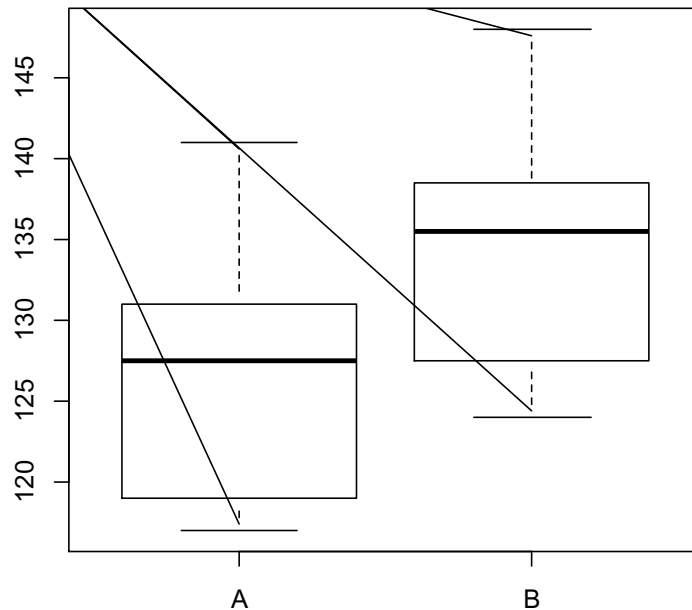
2. The shelf lives of two types of carbonated beverages A and B are to be compared. It is believed that the average shelf life of beverage B is larger than that of the average shelf life of beverage A. Ten bottles of A and 12 bottles of B are tested for shelf lives and the following data are obtained:

Beverage A 119, 127, 117, 141, 128, 117, 130, 132, 131, 122

Beverage B 126, 134, 124, 148, 135, 124, 137, 139, 138, 129, 147, 136

COMPETNECIES – Data Description, Hypothesis Testing, Confidence Intervals

- a) Interpret the following Box-Whisker plot obtained from the experimental data.



- b) State the null hypothesis that you would test. Also state the alternative hypothesis.
c) Stating your assumptions clearly, suggest an appropriate test statistic and calculate its value.
d) Let $pt(q, v)$ denote the area below q in the curve of the t -distribution with v degrees of freedom. Given the following:
 $pt(2.482, 10) = 0.9837842$, $pt(2.482, 20) = 0.9889678$, $pt(2.482, 30) = 0.9905558$,
what is the p -value associated with the computed value of the test statistic? Can you reject the null hypothesis at 5% level of significance? Can you reject it at 1% level?
e) Obtain a 95% confidence interval for the difference in the average shelf lives of beverage B and beverage A.

SOLUTION

- a) The belief that B seems to have a higher mean shelf life than A appears to be true from the plot. Variances seem to be more or less equal. Some hint of skewness in both populations (could be sampling error).
- b) $H_0: \mu_B = \mu_A$, $H_1: \mu_B < \mu_A$ where μ_B and μ_A represent the population means of shelf lives of beverages B and A respectively.
- c) Assumptions: Both populations are normal with the same variance σ^2 . Test statistic: Fisher's t. Computed value = 2.482, df = 20. Details of computation:
- ```
> sA=sd(A) # sample sd of A
> sA
[1] 7.691265
> sB=sd(B) # sample sd of B
> sB
[1] 7.990051
>
> pv = (9*sA^2 + 11*sB^2)/20 # Pooled variance
> pv
[1] 61.7325
>
> num=mean(B) - mean(A) # numerator of Fisher's t
> den = sqrt(1/10 + 1/12)*sqrt(pv) # denominator of Fisher's t
>
> num/den # Computed value
[1] 2.48204
```
- d) p-value =  $1 - p(2.482, 20) = 0.01103$ . Reject null at 5% level.
- e) (1.33, 15.37)

3. In a textile company, an experiment was conducted to study whether K machines were able to provide similar measurements of fiber strength. From the same batch, KxM different samples of fiber were taken, and randomly assigned to the K machines so that each machine measured M similar fiber samples. A one-way analysis of variance (ANOVA) was performed with the experimental data. Answer the following questions.

```
> anova(a)
Response: y (measured fiber strength)
 Df Sum Sq Mean Sq F value
Machine 2 11.1049 ? ?
Residuals 12 ? 0.3541
```

### COMETENCIES – ANOVA

- How many machines were compared in the study, i.e., what is the value of K?
- How many samples of fiber did each machine measure, i.e., what is the value of M?
- Write down the statistical model for the above analysis of variance.
- Fill in the missing values in the ANOVA table.
- Provide an estimate of the residual standard error.
- Let  $pf(q,a,b)$  denote the area below q in the curve of the F-distribution with a,b degrees of freedom. If  $1-pf(10,2,12)=.003$ , what can you conclude from the ANOVA table?

### SOLUTIONS

- K=3
- M=5 (Total DF = 14 = 3x5-1)
- d)

```
> anova(a)
Response: y
 Df Sum Sq Mean Sq F value
factor(x) 2 11.1049 5.5525 15.681
Residuals 12 4.2491 0.3541
```

- Estimate of residual standard error:  $\sqrt{MSE} = 0.595$
- The p-value associated with the F-statistic is much smaller than 0.003. Thus the null hypothesis of equality of the mean measurements made by the machines can be rejected at a level that is less than 0.003.

- 4 Dr. AP has used independent observations from 100 federally funded projects to fit a linear regression model to  $Y = \text{Deficit Spending}$  based on  $X_R = \text{Resources Score}$  and  $X_N = \text{Needs Score}$ . The model fit is

$$Y = a + b_R X_R + b_N X_N + \varepsilon,$$

where “Resources\_Score” and “Needs\_Score” are continuous variables taking on both positive and negative values and  $\varepsilon$  is normally distributed random error with mean 0 and constant variance. Note that a positive value of Y means that the project will have a spending deficit, a negative value means the project will have a spending surplus, and a value of zero means the project will break even.

Dr. AP obtains the following parameter estimates:

| Predictor       | Point Estimate  | Std-Err |
|-----------------|-----------------|---------|
| Intercept       | $\hat{a}$ 10.5  | 5.2     |
| Resources_Score | -5.6            | 1.8     |
| Needs_Score     | $\hat{b}_N$ 7.8 | 3.0,    |

with the following covariances  $\text{Cov}(\hat{a}, \hat{b}_R) = 2.5$ ,  $\text{Cov}(\hat{a}, \hat{b}_N) = -3.2$ ,

$\text{Cov}(\hat{b}_R, \hat{b}_N) = -1.0$ .

#### COMPETENCY - Linear Models

- A. Test the null hypothesis that a project with both Resources\_Score ( $X_R$ ) and Needs\_Score ( $X_N$ ) equaling 0 will have an expected deficit spending of 0 against a two sided alternative at  $p = 0.01$

Point Estimate is  $10.5 + 0(-5.6) + 0(7.8) = 10.5$

Variance of Point Estimate is  $(5.2)^2 + 0(1.8)^2 + 0(3.0)^2 + 2[0(2.5) + 0(-3.2) - 0 * 0 * 1] = (5.2)^2$

..... So the Standard Deviation is 5.2

Thus  $Z = 10.5 / 5.2 = 2.06$ ,  $P > 0.01$  do not reject  $H_0$

- B. Give a 95% confidence interval for the expected deficit spending of a project with a Resources\_Score  $X_R = 1$  and Needs\_Score  $X_N = 1$ .

Point Estimate is  $10.5 + 1(-5.6) + 1(7.8) = 12.7$

Variance of Point Estimate is  $(5.2)^2 + 1(1.8)^2 + 1(3.0)^2 + 2[1(2.5) + 1(-3.2) + 1 * 1 * (-1)]$

$= 27.04 + 3.24 + 9.0 + 5 - 6.4 - 2 = 35.88$

Standard Deviation of Point Estimate is  $\sqrt{35.88} = 5.99$

The 95% Confidence Interval is  $12.7 \pm 1.96(5.99)$  or  $[0.96 \text{ to } 24.44]$

- C. Assume there are two Projects: A. Project-I has a Resources\_Score  $X_R = a$  and Needs\_Score  $X_N = b$  B. Project-II has a Resources\_Score  $X_R = a+1$  and Needs\_Score  $X_N = b+1$  for the same values  $a$  and  $b$ . Calculate the p-value testing the null hypothesis that Project-II has the same expected deficit spending than does Project-I, vs. the alternative that Project-II has a higher expected deficit spending.

Point Estimate for expected change is  $-5.6 + 7.8 = 2.2$

Variance on this point estimate is  $(1.8)^2 + 1(3.0)^2 + 2[-1 * 1 * 1]$   
 $= 3.24 + 9 - 2 = 10.24$

Standard Deviation on the Point estimate is  $\sqrt{10.24} = 3.2$

Looking at the Z-score  $(0 - 2.2) / 3.2 = -0.69$ , there is a 0.25 chance for Z to be less than 0.69 so there is a 0.25 chance that simultaneously increasing the Need\_Score and Resource\_Score by 1 will lower the Expected Deficit.

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## FALL 2018

1. A random sample of 20 adult Busch Bears is obtained from a forest in New Jersey. The following stem & leaf gives the weights in pounds for these animals

5 | 2789  
6 | 12346679  
7 | 1358  
8 | 0112

### COMPETENCIES – Descriptive Statistics, Binomial Distribution

- a) Compute the median weight in the sample \_\_\_\_\_  
 $(26 + 27) / 2 = 26.5$
- b) Compute the interquartile range of weight in the sample \_\_\_\_\_  
 $(21 + 22) / 2 = 20.5$  to  $(35 + 38) / 2 = 36.5$
- c) The median weight of an adult Rutgers-Wombat is known to be 39 pounds. Test the null hypothesis that the median weight of an adult Busch bear is heavier than this against the alternative that it is not at  $P = 0.001$ .

Best answer

$$\left[ \binom{20}{0} + \binom{20}{1} + \binom{20}{2} + \binom{20}{3} + \binom{20}{4} \right] * 0.5^{20} \text{ or}$$

$$2 * [1 + 20 + 190 + 1140 + 4845] * 0.5^{20} \text{ or } 0.0059$$

Do not Reject  $H_0$  at  $P = 0.01$

$$\text{Not so Good Answer } Z = \frac{16 - 10}{\sqrt{0.5 * 0.5 * 20}} = \frac{6}{2.23} = 2.68 \quad P = 0.0076$$

Reject  $H_0$  at  $P = 0.01$

- d) A scientist believes that 25% of adult Busch Bears weigh 11 pounds or less. Is this refuted by the data (explain your answer)?

If this were true the chance to see all 20 adult Busch Bears weigh 12 pounds or more (which is what happened) is  $.75^{20} = 0.003 \dots$ . This is not very likely. So this belief is refuted by the data.

4. Puntious Pilot the Roman who pioneered kicking egg-shaped air inflated balls is building a linear model for force applied to the ball and the distance the ball travels. The model is  $Y = bX + \varepsilon$  where  $X$  is the force Applied,  $Y$  is the distance the ball travels and  $\varepsilon$  is iid error with mean 0 and constant variance  $\sigma^2$ . **NOTE that the intercept of this model is 0.** Puntious has the following data to use to estimate  $b$

|   |   |   |   |   |   |    |    |
|---|---|---|---|---|---|----|----|
| X | 1 | 2 | 3 | 4 | 5 | 6  | 7  |
| Y | 3 | 4 | 5 | 9 | 8 | 20 | 18 |

COMPETENCIES = Linear Models, Least Squares Estimates

- a. Write out the design matrix  $X$

$$[1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7]'$$

- b. Estimate  $b$

$$\hat{b} = \frac{\sum XY}{\sum X^2} = \frac{1 \times 3 + 2 \times 4 + 3 \times 5 + 4 \times 9 + 5 \times 8 + 6 \times 20 + 7 \times 18}{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2} = \frac{348}{140} = 2.49$$

- c. Estimate  $\sigma^2$

|                   |      |      |      |      |       |       |       |
|-------------------|------|------|------|------|-------|-------|-------|
| Y                 | 3    | 4    | 5    | 9    | 8     | 20    | 18    |
| $\hat{Y}$         | 2.49 | 4.98 | 7.47 | 9.96 | 12.45 | 14.94 | 17.43 |
| $(\hat{Y} - Y)^2$ | 0.24 | 0.96 | 6.10 | 0.92 | 19.80 | 25.60 | 0.34  |

$$\text{So } \hat{\sigma}^2 = \frac{\sum (\hat{Y} - Y)^2}{n - 1} = \frac{0.24 + 0.96 + 6.10 + 0.92 + 19.80 + 25.60 + 0.34}{7 - 1} = \frac{53.96}{6} = 8.99$$

- d. What is the variance of  $\hat{b}$  your estimate for  $b$  up above

$$\text{Var}(\hat{b}) = \sum \left( \frac{X}{\sum X^2} \right)^2 \sigma^2 = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2}{(140)^2} = \frac{1}{140} 8.99 = 0.064$$

- e. If a new person kicks the ball with force  $X = 4$ , use the model to give a 95% confidence interval for  $Y$ , how far the ball travels.

$$4\hat{b} \pm t_{6,975} \sqrt{\hat{\sigma}^2 (1 + 4^2 \text{var}(\hat{b}))} = 4\hat{b} \pm t_{6,975} \sqrt{\hat{\sigma}^2 (1 + 4^2 / 140)} = 4 \times 2.49 \pm 2.447 \sqrt{8.99 \times 1.12} \\ = 9.96 \pm 2.447 \times 3.17 = (2.19, 17.73)$$

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## SPRING 2018

1. HIV research scientists are studying the associations of three normally distributed parameters of HIV infection; CD4 count, log-HIV-viral load (LHVL) and P24-Antigen (P24). These are measured on 100 independent subjects and the following sample correlations are observed.

$$r_{\text{CD4,LHVL}} = 0.25$$

$$r_{\text{CD4,P24}} = 0.17$$

$$r_{\text{LHVL,P24}} = 0.35$$

NOTE THE INVERSE TRANSFORMAION OF  $0.5 * \ln\left(\frac{1+x}{1-x}\right)$  is  $\left(\frac{e^{2x}-1}{e^{2x}+1}\right)$

### COMPETENCIES – Statistical properties of correlation coefficients

- a. Use the best test for the null hypothesis that  $\rho_{\text{CD4,P24}} = 0$  against the alternative that  $\rho_{\text{CD4,P24}} > 0$  at  $\alpha = 0.05$

ANSWER test statistic for testing a correlation equals 0 is  $\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$  or  $\frac{0.17\sqrt{100-2}}{\sqrt{1-(0.17)^2}} = \frac{1.68}{0.985} = 1.705 > Z_{1-0.05} = 1.65$  so reject  $H_0$

- b. Give the best 95% confidence interval for  $r_{\text{CD4,LHVL}}$

Do a 95% confidence interval on Fisher's Z transformation  $z_r = 0.5 * \ln\left(\frac{1+r}{1-r}\right)$   
 $= z_r = 0.5 * \ln\left(\frac{1+0.25}{1-0.25}\right) = 0.5(0.51) = 0.255$ . The variance of  $z_r$  is  $\frac{1}{N-3} = \frac{1}{100-3} = \frac{1}{97}$

so the 95% confidence interval for  $0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right)$  is

$$0.255 - 1.96\sqrt{\frac{1}{97}} < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.255 + 1.96\sqrt{\frac{1}{97}}$$

$$0.255 - 1.96(0.101) < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.255 + 1.96(0.101)$$

$$0.057 < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.453$$

Converting this back to a 95% CI for  $\rho$  is

$$\left( \frac{e^{2(0.057)} - 1}{e^{(0.057)} + 1} \right) < \rho < \left( \frac{e^{2(0.453)} - 1}{e^{(0.453)} + 1} \right) \text{ or } \left( \frac{e^{0.114} - 1}{e^{0.114} + 1} \right) < \rho < \left( \frac{e^{0.906} - 1}{e^{0.906} + 1} \right)$$

$$\text{or } \left( \frac{1.12 - 1}{1.12 + 1} \right) < \rho < \left( \frac{2.47 - 1}{2.47 + 1} \right) \text{ or } 0.057 < \rho < 0.424$$

c. Give a point estimate for the partial correlation of LHV L and P24 after adjusting for CD4

$$r_{-LHV L, P24 | CD4} = \frac{r_{-LHV L, P24} - (r_{-CD4, LHV L} \times r_{-CD4, P24})}{\sqrt{1 - (r_{-CD4, LHV L})^2} \sqrt{1 - (r_{-CD4, P24})^2}} = \frac{0.35 - (0.25 \times 0.17)}{\sqrt{1 - (0.25)^2} \sqrt{1 - (0.17)^2}}$$

$$= \frac{0.3075}{\sqrt{0.9375} \sqrt{0.9711}} = 0.322$$

**NOTE - DEPENDING ON THE TEXTBOOKS USED THERE MAY BE SOME VARIATION IN USE OF T DISTRIBUTIONS INSTEAD OF X AND ON THE FORMULAS FOR THE VARIANCE. ALL WILL BE ACCEPTED.**

3. The following data exist on the moisture output from a furnace and the volume if ice forming on the outside tubing over 5 days

|          |   |   |   |   |   |
|----------|---|---|---|---|---|
| Day      | 1 | 2 | 3 | 4 | 5 |
| Moisture | 5 | 1 | 3 | 2 | 4 |
| Ice      | 8 | 1 | 4 | 4 | 8 |

Assume the standard linear model  $Y = a + bX + e$  where Y is ice, X is moisture and e is iid error with mean 0 and a, b are unknown parameters

### COMPETENCIES = Linear Regression

A. Derive the minimum variance unbiased estimates for a and b

$$\bar{X} = (5 + 1 + 3 + 2 + 4) / 5 = 15/5 = 3$$

$$\bar{Y} = (8 + 1 + 4 + 4 + 8) / 5 = 25/5 = 5$$

$$b_{\text{hat}} = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \Sigma(X - \bar{X})^2 \quad (*)$$

|                  |   |    |    |    |   |
|------------------|---|----|----|----|---|
| Day              | 1 | 2  | 3  | 4  | 5 |
| (X - $\bar{X}$ ) | 2 | -2 | 0  | -1 | 1 |
| (Y - $\bar{Y}$ ) | 3 | -4 | -1 | -1 | 3 |

$$\text{So } (*) = (2 \times 3 + -2 \times -4 + 0 \times -1 + -1 \times -1 + 1 \times 3) / (2 \times 2 + -2 \times -2 + 0 \times 0 + -1 \times -1 + 1 \times 1)$$

$$= (6 + 8 + 0 + 1 + 3) / (4 + 4 + 0 + 1 + 1)$$

$$= 18 / 10 = 1.8$$

$$a\_hat = Y\_bar - b\_hat X\_bar = 5 - 3 \times 1.8 = -0.4$$

B. What is an unbiased estimate for the variance of e?

$$\Sigma (Y - Y\_Hat)^2 / n-2 \quad (**)$$

Where  $Y\_hat = a\_hat + b\_hat X$

So for

$$X=1 \quad Y\_hat = -0.4 + 1.8 \times 1 = 1.4$$

$$X=2 \quad Y\_hat = -0.4 + 1.8 \times 2 = 3.2$$

$$X=3 \quad Y\_hat = -0.4 + 1.8 \times 3 = 5$$

$$X=4 \quad Y\_hat = -0.4 + 1.8 \times 4 = 6.8$$

$$X=5 \quad Y\_hat = -0.4 + 1.8 \times 5 = 8.6$$

Or

|       |     |     |   |     |     |
|-------|-----|-----|---|-----|-----|
| Day   | 1   | 2   | 3 | 4   | 5   |
| Y     | 8   | 1   | 4 | 4   | 8   |
| Y-Hat | 8.6 | 1.4 | 5 | 3.2 | 6.8 |

and (\*\*)

$$\begin{aligned}
 &= [(8 - 8.6)^2 + (1 - 1.4)^2 + (4 - 5)^2 + (4 - 3.2)^2 + (8 - 6.8)^2] / (5-1) \\
 &= [0.36 + 0.16 + 1 + 0.64 + 1.96] / 4 \\
 &= 1.20
 \end{aligned}$$

C. It is now proposed that the variance of e is proportional to X in other words  $\text{Var}(e) = c * X$  for some constant c. What approach would you use to derive better estimates for a and b. You are being asked to describe the approach well enough to show full understanding, but do not need to work out the mathematical details to actually derive the estimates.

Use weighted least squares

$$[a\_hat \ b\_hat] = [X' V^{-1} X]^{-1} [X' V^{-1} Y]$$

where

$X'$  is the design matrix

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 3 | 2 | 4 |   |

Y is the outcome matrix

|   |   |   |   |   |  |
|---|---|---|---|---|--|
| 8 | 1 | 4 | 4 | 8 |  |
|---|---|---|---|---|--|

V is a diagonal 5 x 5 matrix with the diagonal elements being

$$\frac{1}{5} \quad 1 \quad \frac{1}{3} \quad \frac{1}{2} \quad \frac{1}{4}$$

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## FALL 2017

2. Dr. Jones is testing a new home delivered education program to increase kindergarten test scores. He will randomly chose 50 independent children to have his program given to them by their parents and then compare the results to 50 other independent children who are not given the program. From normative data, typical education test scored without the intervention have a standard deviation of 15 units and are normally distributed. Higher test scores are better. He will test the null hypothesis that the intervention has not effect on test scores with a two sided  $\alpha = 0.05$ . Let  $\Delta$  represent the true amount of units on average that the intervention improves test score in the population.

COMPETENCIES = Power Estimation,

- A. Assume the intervention has no effect on the standard deviation of a test score and give the power to detect a difference of  $\Delta=14$  under  $H_A$ .

$$(1-\beta) = Z^{-1} \left[ \frac{\Delta}{\sigma \sqrt{2/n}} - Z_{1-\alpha/2} \right] = Z^{-1} \left[ \frac{14}{15 \sqrt{2/50}} - 1.96 \right] = Z^{-1} [4.66 - 1.96] = Z^{-1} [2.70] = 0.9965$$

- B. Now further assume that after the study is done that 21 of the 50 parents who were supposed to give the program to their child did not, so the child had no benefit,
- i. Assuming the 21 parents cannot be identified and all 50 children parents randomized to receive Dr. Jones program must be analyzed together, will power to detect  $\Delta = 14$  under  $H_A$  be more or less than what is calculated above for Part A.? Explain why

Will have lower power than what is calculated in A. as the actual difference will only be  $(29/50) 14 = 8.12$ . Furthermore, the standard deviation of the experimental group will increase.

- ii. If the 21 parents can be identified will excluding their children from the 50 and only comparing the 29 who delivered the program to the 50 controls typically improve power over analyzing al 50 together? Explain why or why not

Should improve power as the difference will once again be  $\Delta=14$  which By removing the 21 “contaminated” observations will offset the reduction in

sample size to 29 in that arm

✂

4. This data exists on the number of hurricanes that have hit Typhoon Island, a place that lies on the equator and thus has hurricanes throughout the year.

NUMBER OF HURRICANES HITTING TYPHOON ISLAND

| Year   | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|--------|------|------|------|------|------|------|------|------|------|------|
| Number | 2    | 5    | 7    | 1    | 4    | 0    | 9    | 3    | 8    | 5    |

COMPETENCIES = Central Limit Theorem, Properties of expectation and variance, Poisson distribution

- A. Assume that the number of hurricanes each year is independent with a common standard deviation, but that the intensity can differ randomly from year to year. Then derive a large sample estimate for the expected number of hurricanes in 2017 and the standard deviation of this estimate.

Normal Distribution Central Limit Theorem

$$\text{Mean} = (2+5+7+1+4+0+9+3+8+5) / 10 = 4.4$$

Variance of Estimated Mean

$$= \{[(2^2 + 5^2 + 7^2 + 1^2 + 4^2 + 0^2 + 9^2 + 3^2 + 8^2 + 5^2) - 10(4.4^2)] / (10-1)\} / 10$$

$$= \{[4 + 25 + 49 + 1 + 16 + 0 + 81 + 9 + 64 + 25] - 193.6\} / 9\} / 10$$

$$= \{(274 - 193.6) / 9\} / 10 = 8.93 / 10 = 0.893$$

$$\text{Std-Err} = \text{SQRT}(0.893) = 0.945$$

- B. Assume that monetary damages from each hurricane is independently distributed including being independent of the number of hurricanes that occur in one year. Use the information on monetary damages from 8 randomly chosen hurricanes below along with your answer to A above to give an estimate for the expected total monetary damages from all hurricanes in 2017

MONETARY DAMAGES FROM 8 RANDOMLY CHOSEN HURRICAINES

| Hurricane | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|-----------|---|---|---|---|---|---|---|---|
| Damages   | 5 | 6 | 4 | 8 | 7 | 6 | 5 | 6 |

For Expected Costs of each Hurricane Based on The Above Table

$$\text{Mean Costs of Damages is } (5+6+4+8+7+6+5+7) / 8 = 5.87$$

Variance of the estimated mean is

$$= \{[(5^2 + 6^2 + 4^2 + 8^2 + 7^2 + 6^2 + 5^2 + 7^2) - 8(5.87^2)] / (8-1)\} / 8$$

$$= \{[(25 + 36 + 16 + 64 + 49 + 36 + 25 + 49) - 275.7] / (8-1)\} / 8$$

$$= [(300 - 275.7) / 7] / 8 = 3.47 / 8 = 0.43$$

Expected Expenditures = Expected Number x Expected Cost Per Event

$$= 4.4 \times 5.87 = 25.83$$

- C. Now assume that the number of hurricanes each year is independent with the same intensity over time and derive a large sample estimate for the expected number of hurricanes in 2017 and the standard deviation of this estimate.

Poisson Distribution

$$\text{Mean} = (2+5+7+1+4+0+9+3+8+5) / 10 = 4.4$$

$$\text{Variance} = 4.4 / 10 = 0.44$$

$$\text{Std-Dev} = \text{SQRT}(0.44) = 0.66$$

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## SPRING 2017

1. THE TOWERS OF THANG-LONG are ceramics that were produced by only two schools in the 14<sup>th</sup> century, the Borneo-cast and the Malacca-cast. It is very difficult to determine which school had manufactured a given ceramic tower but it is known

### Competencies – Bayes Theorem Properties of the Normal Distribution

- height of towers made by the Borneo-cast are distributed  $N(25.00, 10.00)$
  - height of towers made by the Malacca-cast are distributed  $N(30.00, 5.00)$ .
  - also 40% of Thang-Long towers were made by the Borneo cast.
- a. An archeologist discovers a Thang-Long tower that is 27.50 units. Use the above information to estimate the probability that it was made by the Borneo-cast school

$P\{B|B \cup M\} = f\{B\} / [f\{B\} + f\{M\}]$  when M and B are disjoint

$$f(B) = 0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-25)^2}{2 \times 5^2}} \quad f(M) = 0.6 \frac{1}{2\pi 10} e^{-\frac{(27.5-30)^2}{2 \times 10^2}}$$

$$\frac{0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-25)^2}{2 \times 5^2}}}{0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-25)^2}{2 \times 5^2}} + 0.6 \frac{1}{2\pi 10} e^{-\frac{(27.5-30)^2}{2 \times 10^2}}} = \frac{e^{-\frac{6.25}{50}}}{e^{-\frac{6.25}{50}} + 0.75 e^{-\frac{6.25}{200}}} = \frac{e^{-0.125}}{e^{-0.125} + 0.75 e^{-0.03125}} =$$

$$\frac{0.882}{0.882 + 0.75(0.969)} = 0.548$$

- b. An archeologist has 5 Thang-Long towers that were all made by the same school the mean height is 27.50 units. Use this to estimate the probability that it was made by the Borneo-cast school.

$$\text{Now } f(B) = 0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}} \quad \text{and} \quad f(M) = 0.6 \frac{1}{2\pi \frac{10}{\sqrt{5}}} e^{-\frac{(27.5-30)^2}{2 \times \left(\frac{10}{\sqrt{5}}\right)^2}}$$

$$\frac{0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}}}{0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}} + 0.6 \frac{1}{2\pi \frac{10}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{10}{\sqrt{5}}\right)^2}}} = \frac{e^{-\frac{6.25}{10}}}{e^{-\frac{6.25}{10}} + 0.75 e^{-\frac{6.25}{40}}} = \frac{e^{-0.625}}{e^{-0.625} + 0.75 e^{-0.156}} = \frac{0.535}{0.535 + 0.75(0.855)} = 0.455$$

- c. What additional information from the measures of the 5 towers in b. could help to produce a better estimate for the probability that it was made by the Borneo-cast school. Only describe the information not why it could be useful

Knowing the standard deviation of the 5 measures would be useful as Malacca-cast towers standard deviation is smaller so a smaller sample standard deviation would put more evidence towards Malacca-cast and a larger standard deviation would support Borneo-cast.

Or better than that, knowing the value for each of the five observations so that an overall likelihood ratio could be constructed as the product of each of the five LR's

4. THE SNAKES OF THE KALAHARI undergo a long migration during the summer months where many of the snakes die. Dr. Lizard-Beth Taylor (LBT) measured pre-migration weight centigrams (cg) and length in centimeters (Cm) and on 100 snakes plus whether or not the snakes survived the migration.

#### COMPETENCIES – Logistic Regression, Delta Method, Causal Inference

SHE FIRST FITS THIS **LOGISTIC REGRESSION** MODEL FOR SURVIVING MIGRATION THAT USES WEIGHT AS A PREDICTOR

|                         | Estimate | Variance |                                  |
|-------------------------|----------|----------|----------------------------------|
| Intercept ( $\hat{a}$ ) | -0.7     | 0.25     | $Cov(\hat{a}, \hat{b}) = -0.007$ |
| Weight ( $\hat{b}$ )    | 0.2 /cg  | 0.0004   |                                  |

A. Snake A is 2 cg heavier than Snake B. Give an OR and 95% confidence interval for the odds ratio of Snake A surviving the migration Vs. Snake B.

Point Estimate is  $e^{2(0.2)} = e^{0.4} = 1.49$

For 95% CI do 95% CI for b first which is  $0.2 \pm 1.96 * \sqrt{0.0004}$  or  $0.2 \pm 1.96 * 0.02$  or (LL=0.16, UL = 0.24)

Then 95% CI is  $[e^{2*LL} < OR < e^{2*UL}]$  or  $[e^{0.32} < OR < e^{0.48}]$  or  $1.37 < OR < 1.62$

B. Give a point estimate and 95% confidence interval for the probability that a 10 Cg snake survives the migration

$$\text{Point Estimate is } \frac{e^{-0.7+10(0.2)}}{1+e^{-0.7+10(0.2)}} = \frac{e^{1.3}}{1+e^{1.3}} = \frac{3.67}{1+3.67} = 0.79$$

For 95% CI, do 95% CI for  $\alpha + 10$  (a) first

Point Estimate of  $\alpha + 10$  (b) is  $-0.7 + 10(0.2) = 1.3$

$$\begin{aligned} \text{StdDev of } \alpha + 10 (\beta) & \text{ is } \sqrt{\text{Var}(\alpha) + 10^2 (\text{Var}(\beta)) + 2*10(\text{Cov}(\alpha, \beta))} \\ & = \sqrt{0.25 + 100*0.0004 - 2*10*0.007} \\ & = \sqrt{0.15} = 0.39 \end{aligned}$$

So 95% CI for  $\alpha + 10$  ( $\beta$ ) is  $1.3 \pm 1.96 * 0.39$  or (LL=0.53 to UL=2.06)

$$\begin{aligned} \frac{e^{LL}}{1+e^{LL}} < S < \frac{e^{UL}}{1+e^{UL}} & \text{ is a 95\% confidence interval} \\ \frac{e^{0.53}}{1+e^{0.53}} < S < \frac{e^{2.06}}{1+e^{2.06}} & \text{ or } \frac{1.70}{1+1.70} < S < \frac{7.84}{1+7.84} \text{ or } 0.63 < S < 0.88 \end{aligned}$$

SHE THEN FITS TWO MORE MODELS FOR SURVIVING THE MIGRATION WITH [LENGTH] AND [LENGTH + WEIGHT] AS PREDICTORS

|           | Estimate | Variance |  | Estimate  | Variance |        |
|-----------|----------|----------|--|-----------|----------|--------|
| Intercept | 1.3      | 0.32     |  | Intercept | -0.4     | 0.50   |
| Length    | -0.05    | 0.004    |  | Length    | -0.6     | 0.003  |
|           |          |          |  | Weight    | 0.3      | 0.0003 |

C. Use the information on all Three models fitted to determine the best answer for if there is a causal association between length and probability to survive

ANSWER IS NO

From the model with only length, the Z-Value for weight is  $-0.05 / \sqrt{0.004} = -0.05 / 0.2 = -0.25$  which has a two sided P-Value of 0.80

NOTE – The adjusted model using Weight and Length is not a good model to use. Since we know that longer lizards have bigger body volume and thus are heavier. We know that heavier weight is associated with longer survival. It stands to reason that if two lizards have the same weight that the longer lizard is “skinnier” and hence if it is skinnier lizards are more likely to die then length should have a negative coefficient after adjusting for weight.

## FALL 2016

1. People want to compare two plantations for pesticide levels in bananas. They have tested the levels of 6 randomly chosen bunches of bananas each from these two plantations. From each bunch, two randomly chosen bananas are tested. The following pesticide level results are observed for the two bananas within each Bunch

COMPETENCIES = Linear Models / Design of Experiments (Nested Design)

PLANTATION A - Bunch-1 20, 24 Bunch-2 30, 24 Bunch-3 19, 19 Bunch-4 12, 18 Bunch-5 27, 21 Bunch-6 15, 23

PLANTATION B - Bunch-1 20, 14 Bunch-2 20, 24 Bunch-3 16, 14 Bunch-4 17, 13 Bunch-5 17, 15 Bunch-6 19, 17

You can assume the observations are independently normally distributed with equal variance

MODEL  $Y_{ijk} = \mu + b_i I_{\{\text{From Plantation B Effect}\}} + \gamma_j(\text{Plantation}) + \epsilon_{ijk}$

Where  $i = A, B$

$J = 1, \dots, 6$

$k = 1, 2$

NOTE No one should consider the repeated bananas in the same bunch as independent without including the  $\gamma_j$  Bunch Effect

From the data we get  $\bar{Y}_{A1} = 22, \bar{Y}_{A2} = 27, \bar{Y}_{A3} = 19, \bar{Y}_{A4} = 15, \bar{Y}_{A5} = 24, \bar{Y}_{A6} = 19, \bar{Y}_{A..} = 21$   
 $\bar{Y}_{B1} = 17, \bar{Y}_{B2} = 22, \bar{Y}_{B3} = 15, \bar{Y}_{B4} = 14, \bar{Y}_{B5} = 16, \bar{Y}_{B6} = 18, \bar{Y}_{B..} = 17$

The pooled variance of  $\bar{Y}_{B..} Y_{i..}$  is

$$= \frac{(22-21)^2 + (27-21)^2 + (19-21)^2 + (15-21)^2 + (24-21)^2 + (19-21)^2 + (17-17)^2 + (22-17)^2 + (15-17)^2 + (14-17)^2 + (18-17)^2 + (17-17)^2}{(6-1) + (6-1)}$$

$$= \frac{1+36+4+36+9+4+0+25+4+9+1+0}{10} = 12.7$$

So  $\hat{\sigma} = \sqrt{12.7} = 3.56$  with 10 degrees of freedom

A. Give the best 95% confidence interval for the mean level of pesticides in Bananas from Plantation A

$$\bar{Y}_{A..} \pm t_{10,0.975}(\hat{\sigma} / \sqrt{6}) \text{ or } 21 \pm 2.23 \times (3.56 / 2.45) \text{ or } (17.67, 24.24)$$

NOTE – Some people may base the variance only on  $\bar{Y}_{A1}, \bar{Y}_{A2}, \bar{Y}_{A3}, \bar{Y}_{A4}, \bar{Y}_{A5}, \bar{Y}_{A6}$  about  $\bar{Y}_{A..}$  which is 17.4 or std-dev = 4.17 in which case they should also use  $t_{5,0.975} = 2.57$

- B. Give the best statistical test of whether mean pesticide levels differ in Plantation A and B at a two sided  $\alpha=0.05$

$$t_{10df} = \frac{(\bar{Y}_{B..} - \bar{Y}_{A..})}{\hat{\sigma} \sqrt{1/6 + 1/6}} = \frac{(21-17)}{3.56\sqrt{1/3}} = \frac{4}{2.05} = 1.95 \quad p > 0.05 \text{ for 10 d.f.}$$

4. Three types of Fish live in Rutgers pond; Busch-fish, Livingston-fish and Douglas-fish. Drs. HD and DH each recorded fish caught in Rutgers pond from randomly chosen samples.

- Out of a random sample of 120 fish recorded by Dr. HD, 40 were Douglas, 40 were Livingston and 40 were Busch.
- Dr. DH recorded results from a different independent random sample. But he did not record the number of Busch fish Thus out of 90 fish that were either Douglas or Livingston, Dr. HD recorded 60 Livingston and 30 Douglas

Assume that the fish recorded by Drs. HD and DH both random, independent of each other and obtained under the same probabilities to be a Douglas, Livingston or a Busch.

COMPETENCIES = Binomial Distributions, Bayes Theorem and related statistical properties

- A. Use this data from Drs. HD and DH to find the best point estimates for the probability that the species of a fish caught at Rutgers pond were i) Douglas, ii) Livingston and iii) Busch

The only sample that can be used to estimate probability a fish is a Busch is from Dr. HD which is 40/120 = 1/3

The probability to be a Douglas is determined by both samples

$$\begin{aligned} & P\{\text{Not-Busch}\} \quad * \quad P\{\text{Douglas} \mid \text{Not-Busch}\} \\ & \quad 80 / 120 \quad * \quad (40 + 30) / (80 + 90) \\ & \quad \text{From HD} \quad \quad \quad \text{from DH \& HD} \\ = & \quad 2/3 \quad * \quad 70/170 \quad = 14/51 \end{aligned}$$

The probability to be a Livingston is determined by both samples

$$\begin{aligned} & P\{\text{Not-Busch}\} \quad * \quad P\{\text{Livingston} \mid \text{Not-Busch}\} \\ & \quad 80 / 120 \quad * \quad (40 + 60) / (80 + 90) \\ & \quad \text{From HD} \quad \quad \quad \text{from DH \& HD} \end{aligned}$$

$$= \frac{2}{3} * \frac{100}{170} = \frac{20}{51}$$

- B. Give a large sample 95% confidence interval for the probability that the species of a fish caught at Rutgers Pond is a Busch

$$\text{Use } \text{Var}(A * B) = \text{Var}(A) * [E(B)]^2 + \text{Var}(B) * [E(A)]^2$$

Where

A is probability to not be Busch

B is probability to be Douglas | Not-Busch

$$E[A] = 80/120 \quad \text{Var}[A] = [80/120 * 40/120]/120 = 2/1,080$$

$$E[B] = 7/17 \quad \text{Var}[B] = [7/17 * 10/17]/170 = 70/49,130 = 7/4,913$$

$$\text{So } \text{Var}(A*B) = [80/120]^2 * 7/4,913 + [7/17]^2 * 2/1,080 = 0.000947$$

$$\text{Or Std-Dev}(A*B) = (0.000947)^{0.5} = 0.031$$

$$\text{Thus 95\% CI is } 14/51 \pm 1.96 * 0.031 = (0.2745 - 0.0608, 0.2745 + 0.0608) \\ (0.2137, 0.3353)$$

ALTERNATE METHOD USING DELTA METHOD ON LOG TRANSFORMED DATA

$$\text{Var}[\text{Log}(A * B)] = \text{Var}(\text{Log}(A)) + \text{Var}(\text{Log}(B))$$

$$\text{Var}[\text{Log}(A)] = \text{Var}(A)/[E(A)]^2 = 2/1,080 / [80/120]^2 = 9 / 2,160$$

$$\text{Var}[\text{Log}(B)] = \text{Var}(B)/[E(B)]^2 = 7/4,913 / [7/17]^2 = 289 / 34,391$$

$$\text{So } \text{Var}[\text{Log}(A * B)] = 9/2,160 + 289/34,391 = 0.01257$$

$$\text{Std-Dev}[\text{Log}(A * B)] = (0.01257)^{0.5} = 0.112$$

$$\text{So 95\% CI log}(A*B) \text{ is } \text{Log}(14/51) \pm 1.96 * 0.112 = -1.29 \pm 0.22 = (-1.51, -1.07)$$

Exponenting back is  $e^{-1.51}$ ,  $e^{-1.07}$  or (0.22,

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## SPRING 2016

1. An investigator reads a journal article on Total-cholesterol, HDL-cholesterol and LDL-cholesterol from a study of 100 patients where Total-cholesterol = HDL + LDL cholesterol. Assume that for HDL, LDL and Total cholesterol the distributions are iid multivariate normal. The article reports for:

HDL: sample-mean = 145 and p-value for the population mean to be 150 is 0.10

LDL: sample-mean = 48 and p-value for the population mean to be 50 is 0.05

TOTAL: sample-mean = 193 and p-value for the population mean to be 200 is 0.05

All P-values are TWO-SIDED

COMPETENCY = Basic properties of normal distribution and means / covariances

He must derive standard deviations and covariances for HDL and LDL from this information

- A. Work backwards from the p-values for the populations means equaling the specified values based on the observed sample means to obtain the standard deviations for both HDL and LDL in the observations?

For HDL

$$\frac{|150-145|}{\sigma/\sqrt{100}} = Z_{1-0.10/2} \text{ or } \frac{5}{\sigma/\sqrt{100}} = 1.65 \text{ or } \frac{5}{1.65/10} = \sigma \text{ or } \sigma = \frac{50}{1.65} = 30.3$$

For LDL

$$\frac{|50-48|}{\sigma/\sqrt{100}} = Z_{1-0.05/2} \text{ or } \frac{2}{\sigma/\sqrt{100}} = 1.96 \text{ or } \frac{2}{1.96/10} = \sigma \text{ or } \sigma = \frac{20}{1.96} = 10.2$$

- B. What is the covariance of HDL and LDL cholesterol in a single observation?

Since HDL + LDL = TOTAL

$$\text{Var}(\text{TOTAL}) = \text{Var}(\text{HDL}) + \text{Var}(\text{LDL}) + 2*\text{Cov}(\text{HDL}, \text{LDL})$$

$$\begin{aligned} \text{But from A we just saw that } \text{Var}(\text{HDL}) &= (30.3)^2 = 918 \\ \text{Var}(\text{LDL}) &= (10.2)^2 = 104 \end{aligned}$$

To get the Variance of the TOTAL

$$\frac{|200-193|}{\sigma/\sqrt{100}} = Z_{1-0.05/2} \text{ or } \frac{7}{\sigma/\sqrt{100}} = 1.96 \text{ or } \frac{7}{1.96/10} = \sigma \text{ or } \sigma = \frac{70}{1.96} = 35.7$$

$$\text{or } \text{Var}(\text{HDL}) = (35.7)^2 = 1,274$$

$$\begin{aligned}
\text{So } 1,274 &= 918 + 104 + 2 * \text{Cov}(\text{HDL}, \text{HDL}) \\
1,274 - (918 + 104) &= 2 * \text{Cov}(\text{HDL}, \text{HDL}) \\
252 &= 2 * \text{Cov}(\text{HDL}, \text{HDL}) \\
\text{Cov}(\text{HDL}, \text{HDL}) &= 252/2 = 126
\end{aligned}$$

2. Dawn-Holly plants have male and female genders. The time for a single female Dawn-Holly plant grown alone to bloom from birth (i.e. emergence from the ground) in weeks is distributed exponential with  $\lambda=10$ ;  $f(t) = (1/10) e^{-t/10}$ . The time for a single male Dawn-Holly grown alone to bloom is exponential with parameter  $\lambda=15$ ;  $(1/15) e^{-t/15}$ . However, if a female plant is grown paired with (i.e. within 50 feet of) a male plant the time to blooming for both plants is independent except that once the male plant blooms, the pollen released by the male triggers the female plant to immediately bloom if the female plant did not bloom first. Thus when a male and female Dolly-Holly are paired together, the time for the female plant is  $\min(Y, Z)$  where  $Y \sim \exp(10)$  and  $Z \sim \exp(15)$ .

COMPETNECY = Properties of exponential and normal approximation

- A. Compared to growing a female Dolly-Holly alone, by how much time on average does pairing the female plant with a male Dolly-Holly plant reduce the expected time to bloom for a female Dolly-Holly?

The mean of the time to blooming for a female plant that is not paired is  $1/(1/10) = 10$  weeks.

When paired with the main plant, the CDF if time to bloom for the female plant is  $e^{-t/10} \times e^{-t/15} = e^{-5/30} = e^{-1/6}$ . So the mean time to bloom is now 6 weeks.

This means that adding the male plant has reduced the time to bloom on average by 4 weeks =  $10 - 6$  weeks

- B. If  $N=100$  independent pairs of female and male Dolly-Holly plants are grown, Let  $X$  be then number of pairs where the male Dolly-Holly bloomed first and stimulated the female plant to bloom. What is the probability that  $X \geq 50$ .

For each pair, the probability the male blooms first is  $4/10$  derived as follows. The probability the male blooms first is

$$\begin{aligned}
\int_0^{\infty} f(\text{male blooms at } x) (\text{Pr ob female Not Bloomed By } X) &= \\
\int_0^{\infty} (1/15)e^{-t/10} e^{-t/15} dt &= \int_0^{\infty} (1/15)e^{-(t/10+t/15)} dt = \int_0^{\infty} (1/15)e^{-(t/6)} dt = \left|_0^{\infty} -(6/15)e^{-(t/6)} \right. \\
&= 6/15 = 0.4
\end{aligned}$$

So using the normal approximation to the binomial, the probability to observe 50 or more pairs (0.5 of the 100 pairs) in which the male blooms first is 1 minus the CDF of  $Z$  where

$$Z = \frac{0.5-0.4}{\sqrt{(0.4)(1-0.4)/100}} = \frac{0.1}{0.0489} = 2.04 \text{ and } Z^{-1}(2.04) = 0.9793 \text{ so the answer is}$$

$$1 - 0.9793 = 0.02$$

## FALL 2015

1. Dr. Jones obtained least squares linear regression estimates for parameters of this linear model with an interaction term  $Y = a + b_1 X_1 + b_2 X_2 + b_{12} X_1 X_2 + \epsilon$  using 94 adults; where  $Y$  is hematocrit,  $X_1$  is sex (0=female, 1=male)  $X_2$  is [age in years – 30];  $a$ ,  $b_1$ ,  $b_2$ , and  $b_{12}$  are unknown constants; and  $\epsilon$  is normally distributed error with mean 0 and unknown variance  $\sigma^2$  and obtained the following estimates. You can assume that all of these previous assumptions are valid when you answer the questions.

| PARAMETER | ESTIMATE | STD-ERROR |
|-----------|----------|-----------|
| $a$       | 40       | 1.1       |
| $b_1$     | -0.1     | 0.025     |
| $b_2$     | -2       | 0.5       |
| $b_{12}$  | -0.5     | 0.025     |

and  $\hat{\sigma}^2 = 10$

- A. Give a 95% confidence interval for  $\sigma^2$  or if you cannot answer the question from the information given explain what additional information you would need to answer the question

### COMPETENCY = LINEAR MODELS

$(\hat{\sigma}^2 / \sigma^2) \times 90$  has a chisquare distribution with  $94-4 = 90$  d.f.

So  $(\chi^2_{(94,0.975)} / 90)\hat{\sigma}^2 < \sigma^2 < (\chi^2_{(94,0.025)} / 90)\hat{\sigma}^2$  is a balanced 95% confidence

interval for  $\sigma^2$  where  $\chi^2_{(94,0.975)} = 65.647 / 90 = 0.73$  and

$\chi^2_{(94,0.025)} / 90 = 118.136 / 90 = 1.31$

thus  $0.73 * 10 < \sigma^2 < 1.31 * 10$

$7.3 < \sigma^2 < 13.1$  is a balanced 95% CI for  $\sigma^2$

- B. Give a 95% confidence interval for the mean hemoglobin of 30 year old women or if you cannot answer the question from the information given explain what additional information you would need to answer the question

Start By Getting the Expected Value First:

For a 30 years old woman,  $X_1 = X_2 = 0$  so  $E[Y] = a + b_1 X_1 + b_2 X_2 + b_{12} X_1 X_2$

$E[Y] = a + b_1 0 + b_2 0 + b_{12} 0 = a$

And Now you can see that the 95% CI for  $a$  is the 95% confidence interval for  $E[Y]$  of a 30 year old woman .... Since  $df = 90$  which is  $> 60$  you can use Z statistic and

$\hat{a} - 1.96(std - err(\hat{a})) < a < \hat{a} + 1.96(std - err(\hat{a}))$

$40 - 1.96(1.1) < a < 40 + 1.96(1.1)$  or  $40 - 2.16 < a < 40 + 2.16$  or

$(37.84, 42.16)$  is the 95% confidence interval for the expected hematocrit of a 30 year old woman

4. Dr. Lee developed The Dooku-Test for infection by the Dooku virus.
- I. He tests 1,000 persons who truly have the Dooku virus The Dooku-Test
    - correctly identifies that 990 have the Dooku virus
    - falsely identifies that 10 do not have the Dooku virus
  - II. He then tests 1,000 persons who don't have the Dooku virus The Dooku-Test
    - correctly identifies that 990 do not have the Dooku virus
    - falsely identifies that 10 do have the Dooku virus

Remember that the numbers in I and II are sample estimates

Suppose in the United States that exactly 1/1000 of the persons (0.001) are infected with the Dooku virus which means 0.999 are not. You should take this number to be known and not an estimate

COMPETENCIES = Applied Probability, Normal Approximation to Binomial, Bayes Theorem

- A. Give the best point estimates for
- a. probability that a US person is both infected with the Dooku virus and would test positive for Dooku by the Dooku-Test.

ANSWER

$$P\{\text{infected}\} \times P\{\text{test positive} \mid \text{infected}\} \\ 0.001 \quad * \quad 0.99 \quad \quad \quad = 0.00099$$

- b. Give the best point estimated for the probability that a US person is both not infected with the Dooku virus and would test positive for Dooku by the Dooku Test.

ANSWER

$$P\{\text{Not infected}\} \times P\{\text{test positive} \mid \text{not infected}\} \\ (1-0.001) \quad * \quad 0.01 \quad \quad \quad = 0.0099$$

- B. Give a large sample 95% confidence interval for the point estimate in A-a. up above

It is known that 0.001 are infected so with Dooku this is a constant. The 0.99 probability for a person infected with Dooku to test positive by the Dooku test is a point estimate. The 95% large sample CI

$$\text{is } 0.99 \pm 1.96 \sqrt{\frac{0.01 * 0.99}{1,000}} \text{ or } 0.99 \pm 0.003 \text{ or } 0.987 - 0.993.$$

Since the probability to be infected with the Dooku virus was given as a constant, we just use it to multiply these limits

Thus  $[0.001 \times 0.987]$  to  $[0.001 \times 0.993]$  or **0.000987 – 0.000993** is a 95% confidence interval for a US person to be infected with Dooku and to test positive

- C. If a US person tests positive for the Dooku virus, give the best point estimate for the probability they are really infected with that virus.

$$\text{From Bayes Theorem } P\{A \mid (A \cup B)\} = \frac{P\{A\}}{P\{A\} + P\{B\} - P\{A \cap B\}}$$

$$P\{A\} + P\{B\} - P\{A \cap B\} \quad \text{where}$$

A = Person is Truly positive and tests positive  $P\{A\} = 0.0099$

B = Person is Truly negative and tests positive  $P\{B\} = 0.00099$

$P\{A \cap B\} = 0$  since a person can't be both truly positive and Truly negative at the same time

$$\text{So } P\{A \mid (A \cup B)\} = \frac{0.00099}{0.00099 + 0.099} = \frac{1}{11} = 0.0909$$

## FALL 2014

- Below are data on numbers of AIDS Cases, number of AIDS Cases caused by PCP and KS for North American cities. (Note AIDS Being Caused by PCP and by KS is mutually exclusive).

| CITY       | AIDS CASES |               |              |
|------------|------------|---------------|--------------|
|            | TOTAL      | CAUSED BY PCP | CAUSED BY KS |
| Des Moines | 200        | 30            | 30           |
| El Paso    | 200        | 20            | 50           |

Let  $p_D$  be the portion of all AIDS cases in Des Moines caused by PCP  
 $p_E$  be the portion of all AIDS cases in El Paso caused by PCP

$k_D$  be the portion of all AIDS cases in Des Moines caused by KS  
 $k_E$  be the portion of all AIDS cases in El Paso caused by KS

### COMPETENCIES

Two BY Two Table Word Problem – Question a

Conditional Chisquare Approximation Word Problem – Question b

- Dr. LP believes that  $p_E = p_D$ . Give a large sample test of this alternative hypothesis against the appropriate null at  $\alpha=0.05$

### TWO BY TWO TABLE (WITHOUT CONTINUITY CORRECTION)

|            | AIDS - PCP+ | AIDS - PCP+ | TOTAL |
|------------|-------------|-------------|-------|
| Des Moines | 30          | 170         | 200   |
| El Paso    | 20          | 180         | 200   |
| TOTAL      | 50          | 350         | 400   |

$$\frac{(180 * 30 - 170 * 20)^2 400}{50 * 350 * 200 * 200} = 2.28 \quad P > 0.05 \text{ two sided } H_0$$

- Dr. PL believes  $k_D = 2p_D$ . Give a large sample test of this alternative hypothesis against the appropriate null at  $\alpha=0.05$

### CONDITIONAL TWO BY TWO TABLE (WITHOUT CONTINUITY CORRECTION)

|            | AIDS - PCP | AIDS - KS | TOTAL |
|------------|------------|-----------|-------|
| Des Moines | 30         | 30        | 60    |

Conditional on 30 PCP + KS cases = 60 total in Des Moines, under the null hypothesis the expected number of PCP and KS cases are 20 and 40 each

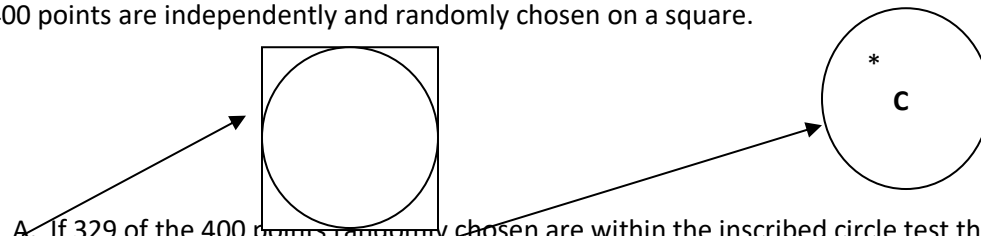
$$\sum \frac{(O - E)^2}{E} = \frac{(30 - 20)^2}{20} + \frac{(30 - 40)^2}{40} = 5 + 2.5 = 7.5$$

Compare this to 3.84 the upper (1-0.05) level of a  $\chi^2_{1df}$  and reject  $H_0$

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## SPRING 2014

2. 400 points are independently and randomly chosen on a square.



A. If 329 of the 400 points randomly chosen are within the inscribed circle test the null hypothesis of the probability to fall within the circle is random versus the alternative that it is greater than by chance alone relative to area of the circle at  $\alpha = 0.05$ .

B. Now consider only the 329 independent points inside the inscribed circle. Suppose the radius of this circle is 1 and the sample mean distance of a point from the center of the circle (C) is  $\bar{X} = 0.65$  with a variance of  $S^2 = 0.056$  (you can use this variance to estimate the variance for the population). Give a large sample test of the two sided null hypothesis on whether or not the points are closer to the center of the circle than they should be by chance alone if the points were randomly distributed inside the circle.

### COMPETENCIES

Single Proportion Hypothesis Normal Approximation Word Problem - Question A  
Using Calculus to Find Expected Value and Normal Approximation – Question B

#### 2. ANSWER TO A

i). If  $r$  is the radius of the inscribed circle, then  $2r$  is the height and width of the square. The area of the inscribed circle is  $\pi r^2$  and the area of the square is  $4r^2$ . So by chance alone, the probability to be inside the circle is  $\pi r^2 / 4r^2 = \pi/4 = 0.785$ . This means if 400 points are independently and randomly chosen, and expected 0.785 will be inside the circle and the variance of this observed portion is  $0.785 * 0.215 / 400$ .

ii) In 400 tries 329 or 0.8225 of the points were inside the circle.

iii) Assume the Null Hypothesis and do a Z test

$$Z = \frac{0.8225 - 0.785}{\sqrt{0.785 * 0.215 / 400}} = \frac{0.0375}{0.0205} = 1.83$$

Since  $1.83 > 1.65 = Z_{0.95}$  reject  $H_0$  for  $H_a$  that there is a greater than expected (by relative area) chance for a point to be inside the inscribed circle

#### ANSWER TO B

- i) The expected distance of a point from the Center of the Circle is:

$$\frac{\int_0^1 x(2\pi x)dx}{\pi} = \frac{\int_0^1 2\pi x^2 dx}{\pi} = \frac{\left|_0^1 (2/3)\pi x^3\right.}{\pi} = 0.667$$

ii) Using The Z-Test,  $Z = \frac{0.65 - 0.667}{\sqrt{0.056/329}} = \frac{-0.0167}{0.0130} = -1.23$

- iii) Since  $|-1.23| < 1.96$  do not reject  $H_0$

NOTE - Some students may figure out by using calculus to find  $E[x^2]$  that if the distribution of points is random, then the **variance** of the distance from the center of the circle is  $\frac{1}{2} - \frac{4}{9} = 0.0555555\dots$  so the “standard deviation” being 0.056 is likely a misprint. If so and a student used 0.056 as the variance rather than the standard deviation they would get .... And in fact would probably get most to full credit even if 0.056 did turn out to be the standard deviation, but the student showed their calculations as to why they thought it was the variance.

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## FALL 2013

1. Assume that birthdays are iid distributed across 365 days of the year (exclude Feb 29<sup>th</sup> on leap year). We have two groups of people

G1 – Has 10 Persons

G2 – Has 10 different persons

### Applications of Probability

- A. What is the probability that no two persons in G1 were born on the same day of the year?

$$364/365 \times 363/365 \times \dots \times 356/365 = 0.883$$

- B. Assume that no two persons in G1 have the same birthdate, then what is the probability that none of the persons in G2 were born on the same day as a person in G1?

$$(355/365)^{10} = 0.757$$

- C. Assume again that no two persons in G1 have the same birthdate, what is the probability that none of the 20 persons in G1 and G2 combined have the same birthdate?

$$355/365 \times 354/365 \times \dots \times 346/365 = 0.667$$

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## SPRING 2013

3. A Statistics Professor believes that adding videos of statisticians consulting on applied problems to her standard instructional material will help students solve problems. She designs an experiment to find evidence to support this hypothesis. A total of 167 students are given a test prior to receiving her class (**the “pretest”**) to see how they approach applied statistics problems. The pretest is graded Pass or Fail. The students are then randomly divided into two groups: The Experimental Group with 106 students, and the Control Group with 61 students. Over the next 2 weeks, the Experimental group are shown videos plus her standard instructional material, while the Control group receives only her standard instructional material. After this is done, all students take the same Pass/Fail test again on solving applied statistics problems (**the “post-test”**). You can assume that all students are independent. The following are the combined Pretest Posttest outcomes:

| Group        | Pretest | Posttest | Number of subjects |
|--------------|---------|----------|--------------------|
| Experimental | Fail    | Fail     | 37                 |
| Experimental | Fail    | Pass     | 29                 |
| Experimental | Pass    | Fail     | 11                 |
| Experimental | Pass    | Pass     | 29                 |
| Control      | Fail    | Fail     | 26                 |
| Control      | Fail    | Pass     | 9                  |
| Control      | Pass    | Fail     | 7                  |
| Control      | Pass    | Pass     | 19                 |

- Using only the data for the experimental group is there statistically significant evidence of an improvement between Pretest and Posttest performance. What is the Null Hypothesis; the Alternative Hypothesis, and show the details for the test used.
- Using only the data for the control group is there statistically significant evidence of an improvement between the pretest and posttest?
- Now combine the analyses for the experimental and control groups to see if there is statistically significant evidence that adding the video to the instruction was helpful.

### COMPETENCY

McNemars Discordant Pairs

Two by Two Table Word Problem

### ANSWER

$H_0$ : Probability to Pass post video is the same as the probability to pass pre-video

$H_A$ : Probability to Pass post video is higher than probability to pass pre-video

a)\_ Two by Two Table of Matched Responses in Experimental Group

| Pretest Result | Posttest Result |      |
|----------------|-----------------|------|
|                | Fail            | Pass |
| Fail           | 37              | 29   |

|      |           |           |
|------|-----------|-----------|
| Pass | <b>11</b> | <b>29</b> |
|------|-----------|-----------|

Focus on the discordant pairs  $11 + 29 = 40$ . If  $H_0$  is true, conditional on 40 discordant pairs, the expected number who fail pre and pass post = number who pass pretest and fail post test =  $40 / 2 = 20$ . If  $H_A$  true, the number who pass post and fail pre is larger.

Test Statistic is  $\sum \frac{(E-O)^2}{E} = \frac{(29-20)^2}{20} + \frac{(11-20)^2}{20} = 8.1$  Compare to  $\chi^2_1(1-0.05) = 3.84$ .  $P < 0.05$

b)\_ Two by Two Table of Matched Responses in Control Group

$H_0$ : Probability to Pass post video is the same as the probability to pass prevideo

$H_A$ : Probability to Pass post video is higher than probability to pass prevideo

| Pretest Result | Posttest Result |           |
|----------------|-----------------|-----------|
|                | Fail            | Pass      |
| Fail           | <b>26</b>       | <b>9</b>  |
| Pass           | <b>7</b>        | <b>19</b> |

Focus on the discordant pairs  $7 + 9 = 16$ . If  $H_0$  is true, conditional on 8 discordant pairs, the expected number who fail pre and pass post = number who pass pretest and fail posttest =  $(7+9)/2 = 8$ . If  $H_A$  true, the number who pass post and fail pre is larger.

Test Statistic is  $\sum \frac{(E-O)^2}{E} = \frac{(9-8)^2}{8} + \frac{(7-8)^2}{8} = 0.25$  Compare to  $\chi^2_1(1-0.05) = 3.84$ .  $P > 0.05$

c) There are some potential problems with any answer to this that would be easy enough to complete on the exam. I believe this answer which tests if the matched OR for discordant pairs was the same for each treatment arm would get full credit. Comparative two by two table of discordant results in experimental versus control arm.

$H_0$ : Ratio of Probability to Pass-post / Pass-pre video in the discordant pairs is the same for both Experimental and Control arms

$H_0$ : Ratio of Probability to Pass-post / Pass-pre video in the discordant pairs is greater for Experimental arm

| Pretest Result | Posttest Result    |                   |
|----------------|--------------------|-------------------|
|                | Fail-Pre Pass-Post | Pas-Pre Fail-Post |
| Experimental   | <b>29</b>          | <b>11</b>         |
| Control        | <b>9</b>           | <b>7</b>          |
|                | <b>38</b>          | <b>18</b>         |

**56**

**Test of Independence**  $\frac{(29*7-9*11)^2 56}{38*18*40*16} = 1.38$  Compare to  $\chi^2_1(1-0.05) = 3.84$ .  $P > 0.05$

So the 95% CI is  $4.79 < X < 26.24$